

Phase 1: Invoice Platform Workflow

Overview

The invoice platform contains four portals: User, Officer (Stage 1, Stage 2, Stage 3 using role-based tokens), Admin (multiple supported), and Super Admin (multiple supported; currently one).

Platform Components

- **User Portal** – Used to create, manage, and submit invoices.
- **Officer Portal** – Shared by **Stage 1, Stage 2, and Stage 3 officers**. Each officer accesses the portal using role-specific authentication tokens.
- **Admin Portal** – Supports multiple administrators responsible for managing users, officers, and operational tasks.
- **Super Admin Portal** – Provides full system administration. The platform currently has one Super Admin but supports multiple Super Admin accounts.

User Registration & Verification

Before a user can create invoices, they must first register for an account.

Current Verification Process

1. The user creates an account and submits the required business information.
2. During registration, the user must provide their Gem Dealer File Number along with other required registration details.
3. The registration request is submitted for verification.
4. An administrator verifies the submitted information against the official records.
5. This verification process typically takes up to one working day.
6. Once approved, the user's account is activated, allowing access to the invoice platform and all invoice-related features.

Invoice Creation Flow

1. A user logs into User Portal and creates a new invoice.
2. The user selects a Category and then a Subcategory. The available subcategories depend on the selected category.
3. Once both are selected, the system automatically loads the assigned invoice template. Each template is linked to at least one category or subcategory.
4. The user completes the invoice by entering the required information.
5. Before submission, the user can preview the generated invoice.
6. The invoice can either be:
 - **Saved as Draft** – allowing the user to edit and submit it later.
 - **Submitted** – initiating the approval workflow.

Approval Workflow

Once an invoice is submitted, it progresses through a **three-stage officer approval process**.

Stage 1

- The invoice is assigned to a **Stage 1 Officer**.
- The officer reviews and, if necessary, edits the invoice.
- Any edits made by the officer are sent back to the user for approval.
- Only after the user accepts the changes can the officer continue processing the invoice.
- The officer then forwards the invoice to **Stage 2**.

Stage 2

- A **Stage 2 Officer** performs the same review and editing process.
- If changes are made, the invoice is returned to the user for approval.
- Once approved by the user, the officer forwards the invoice to **Stage 3**.

Stage 3

- A **Stage 3 Officer** performs the final review.
- Any modifications again require user approval before processing can continue.
- Once the Stage 3 Officer approves the invoice, the approval workflow is complete.

Future Enhancement – Automated Dealer Verification

The current manual verification process can be eliminated by integrating the platform with the official **Gem Dealer Database**.

With this integration:

- The system will automatically validate the submitted **Gem Dealer File Number** during user registration.
- If the dealer information is successfully verified, the user's account can be activated immediately without administrator intervention.
- This will significantly reduce registration time while maintaining data accuracy, security, and regulatory compliance.

If the relevant systems or digital services already exist, they can be integrated with the platform through APIs or other secure data exchange mechanisms. Where necessary, these existing systems can be extended or customized to support the platform's verification workflow.

As a future enhancement, the same integration can be expanded into a **fully online Gem Dealer License application and renewal platform**. This would allow new applicants to submit license applications digitally, enable existing dealers to renew or update their licenses online

Phase 1: Digital Letter Tracking & QR Verification System (ERP)

Project Overview

This proposal outlines the first phase of the ERP system, focusing on digitizing the organization's official letter issuance and tracking process. The objective is to provide a secure, traceable, and efficient method of managing official documents while supporting both digital and hard-copy workflows.

Phase 1 Scope

1. Letter Creation

The User will be able to create an official letter by entering the required details, including:

- Upload official seal
- Upload digital signature
- Recipient details
- Subject/Purpose of the letter
- Department
- Date
- Reference number
- Additional remarks (optional)

The system will generate a unique record for every letter.

2. QR Code Generation

Once the letter is finalized, the system will generate a unique QR code linked to the letter record.

The QR code can be:

- Printed directly on digitally generated letters.
- Attached to hard-copy documents as a sticker or printed label.

The QR code will **not** contain the document itself. Instead, it will securely reference the letter information stored in the system.

3. QR Verification

When the QR code is scanned, the system will display:

- Letter reference number
- Sender
- Recipient
- Date issued
- Department
- Subject/Purpose
- Current status
- Verification status

This allows recipients to verify that the physical document is genuine and officially issued.

4. Letter Movement Tracking

The system will track the movement of each letter throughout the organization.

Examples include:

- Created
- Sent
- Received
- Forwarded
- Pending
- Completed

Each movement will include:

- Department
- User
- Date & Time
- Remarks (optional)

5. Super Admin Dashboard

The Super Administrator will be able to:

- View all issued letters
- Monitor letter movement between departments
- View complete movement history
- Search and filter letters
- View verification history
- Manage departments and users

6. Department Support

The system will support multiple departments, including but not limited to:

- HR
- Legal
- IT
- Export
- Finance
- Administration
- Additional departments as required

Security Considerations

Initial Approach

For the initial prototype, a standard QR code will be generated that can be scanned using any QR scanner.

While this enables quick implementation, it introduces security concerns because anyone with the QR code could potentially access the verification page.

Recommended Approach

For production deployment, we recommend developing a dedicated mobile application for authorized personnel.

Benefits include:

- User authentication before viewing document details
- Role-based access control
- Secure document verification
- Scan history logging
- Better protection against unauthorized access
- Enhanced tracking and audit capabilities

This approach provides significantly stronger security than a publicly accessible QR verification page.

Process Study Requirement

Before beginning full system development, it is recommended that the existing manual document workflow be reviewed.

Understanding the current approval, dispatch, routing, and record-keeping process will allow the ERP system to accurately reflect existing operations while identifying opportunities for automation and improvement.

Expected Deliverables (Phase 1)

- Letter creation module
- Signature and seal management
- QR code generation
- QR verification portal
- Letter movement tracking
- Super Admin dashboard
- Department management
- Basic reporting and search functionality

Future Enhancements

- Mobile application for secure QR verification
- Digital approval workflow
- Notification system
- Document archive
- Analytics dashboard
- Integration with the complete ERP platform
- Multi-level approval workflows
- Audit logs and compliance reporting

Conclusion

Phase 1 establishes the foundation for a secure document management and tracking system. By combining QR-based verification with centralized tracking, the organization will gain improved transparency, accountability, and operational efficiency. Following an assessment of the current manual process, the system can be refined to align closely with existing business workflows and expanded into a comprehensive ERP solution.